

10a-sentiment-analysis-with-text-mining-with-r

Introduction

This assignment reproduces the sentiment analysis example from Chapter 2 of *Text Mining with R* (Silge & Robinson, 2017) and extends it with a new corpus and an additional sentiment lexicon.

Citation: Silge, J., & Robinson, D. (2017). *Text Mining with R: A Tidy Approach*. O'Reilly Media. Chapter 2: Sentiment Analysis with Tidy Data. Available at <https://www.tidytextmining.com/sentiment>. The code in Part 1 is adapted directly from that chapter.

Setup

```
library(tidyverse)
library(tidytext)
library(janeaustenr)
library(gutenbergr)
```

Part 1: Reproducing the Base Example

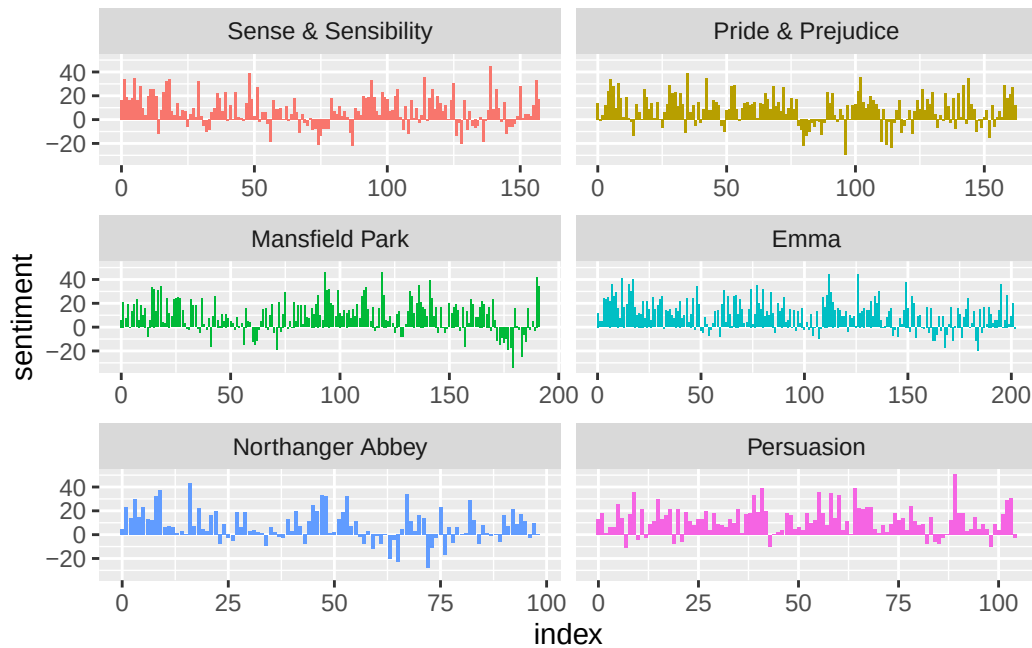
The Chapter 2 example tokenizes Jane Austen's novels, joins each token with the **bing** lexicon, and plots net sentiment (positive minus negative) across 80-line sections of each book.

```
tidy_books <- austen_books() %>%
  group_by(book) %>%
  mutate(
    linenumber = row_number(),
    chapter = cumsum(str_detect(text,
                                regex("^chapter [\\divxlc]",
                                       ignore_case = TRUE)))) %>%
  ungroup() %>%
  unnest_tokens(word, text)

jane_austen_sentiment <- tidy_books %>%
  inner_join(get_sentiments("bing")) %>%
```

```
count(book, index = linenumber %/% 80, sentiment) %>%
pivot_wider(names_from = sentiment, values_from = n, values_fill = 0) %>%
mutate(sentiment = positive - negative)

ggplot(jane_austen_sentiment, aes(index, sentiment, fill = book)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~book, ncol = 2, scales = "free_x")
```



This reproduces Figure 2.2 of the book.

Part 2: Extension

New Corpus

We pull *The Adventures of Sherlock Holmes* (Arthur Conan Doyle, 1892) from Project Gutenberg (ID 1661).

```
holmes <- gutenbergl_download(1661, mirror = "https://gutenberg.pglafl.org/")

tidy_holmes <- holmes |>
  mutate(linenumber = row_number()) |>
  unnest_tokens(word, text)
```

Additional Lexicon: Loughran-McDonald

The Loughran-McDonald lexicon (Loughran & McDonald, 2011) was developed for financial and regulatory text (10-K filings, earnings releases). It is distributed with `tidytext` but is *not* used in the Chapter 2 example, so we treat it as our additional lexicon. It classifies words into six categories: `positive`, `negative`, `litigious`, `uncertainty`, `constraining`, and `superfluous`. For comparability we keep only the positive/negative categories.

Citation: Loughran, T., & McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1), 35-65.

Comparing Four Lexicons on Sherlock Holmes

We score net sentiment over 80-line sections using `bing`, `nrc`, `afinn`, and `loughran`. For `afinn` we binarize the integer scores (non-negative = positive) so all four are on the same footing.

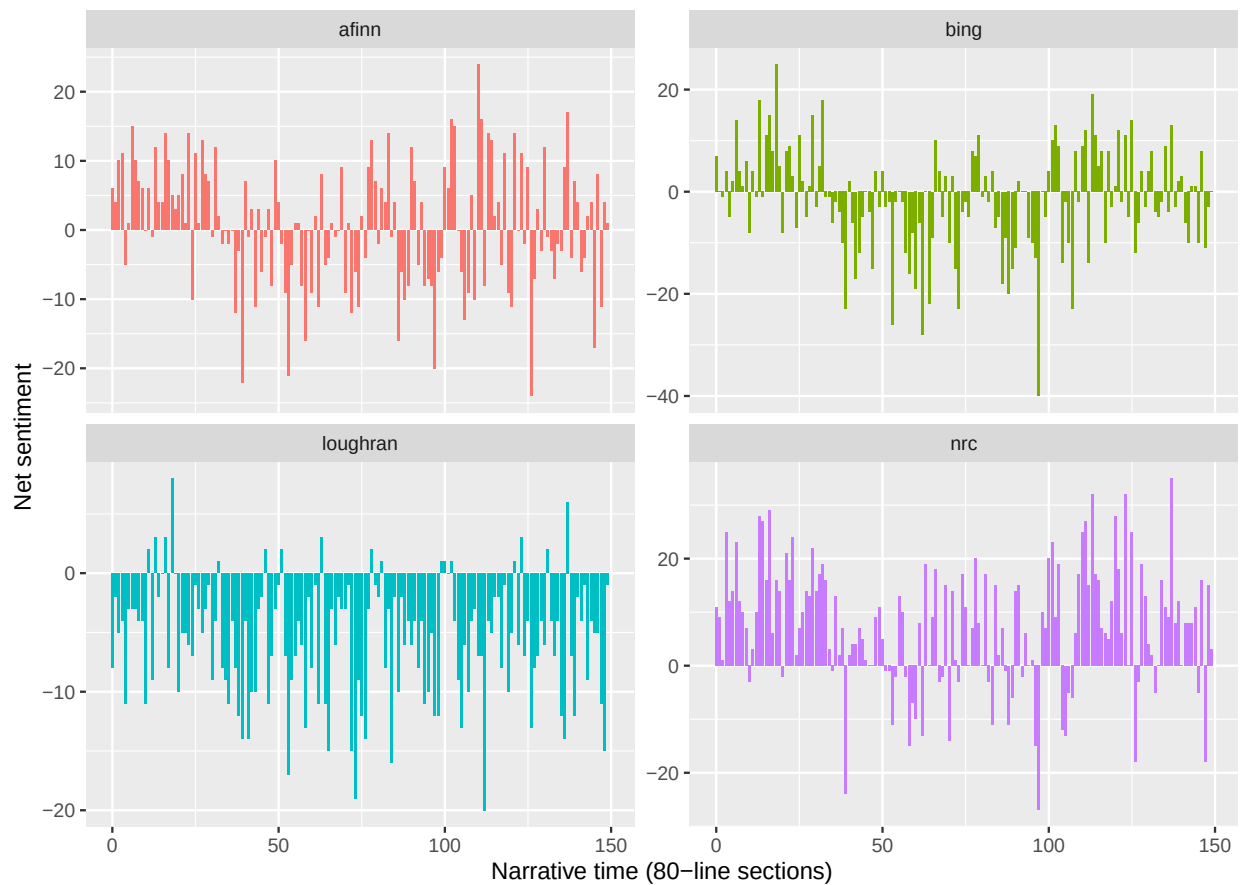
```
binary <- function(lex) {
  s <- get_sentiments(lex)
  if (lex == "afinn") {
    s |> transmute(word, sentiment = if_else(value >= 0, "positive", "negative"))
  } else {
    s |> filter(sentiment %in% c("positive", "negative"))
  }
}

score <- function(lex) {
  tidy_holmes |>
    inner_join(binary(lex), by = "word", relationship = "many-to-many") |>
    count(index = linenumbers %/% 80, sentiment) |>
    pivot_wider(names_from = sentiment, values_from = n, values_fill = 0) |>
    mutate(sentiment = positive - negative, lexicon = lex)
}

holmes_compare <- bind_rows(
  score("bing"),
  score("nrc"),
  score("afinn"),
  score("loughran")
)

ggplot(holmes_compare, aes(index, sentiment, fill = lexicon)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~lexicon, ncol = 2, scales = "free_y") +
  labs(title = "Sentiment through 'The Adventures of Sherlock Holmes'",
       x = "Narrative time (80-line sections)",
       y = "Net sentiment")
```

Sentiment through 'The Adventures of Sherlock Holmes'



Summary Statistics

```
holmes_compare |>
  group_by(lexicon) |>
  summarise(
    mean_net = round(mean(sentiment), 2),
    sd_net   = round(sd(sentiment), 2),
    pct_neg  = round(mean(sentiment < 0) * 100, 1)
  )
```

```
# A tibble: 4 x 4
  lexicon mean_net sd_net pct_neg
  <chr>     <dbl>  <dbl>  <dbl>
1 afinn      0.67   8.84   43.3
2 bing     -1.52  10.1   53.3
3 loughran -5.63   5.07   86.7
4 nrc       7.29  11.5   22.7
```

How Do the Results Differ?

- **nrc** shows a strong positive bias because its positive category contains many common everyday words. Holmes looks overwhelmingly positive under **nrc**, even in chapters built around murder and blackmail.
- **bing** is more balanced and picks up the darker plot points that **nrc** washes out. This matches what Silge and Robinson observe in Section 2.3.
- **afinn**, once binarized, looks similar to **bing** in shape. The raw integer weights (dropped here for comparability) would make it more sensitive to a single strongly negative word per section.
- **loughran** was trained on corporate filings, so most of its vocabulary (**litigation**, **restructuring**, **impairment**) does not appear in a Victorian detective story. Its signal is sparse and mostly negative, driven by crime-adjacent overlap words like **crime**, **guilty**, and **suspicious**. This illustrates a key point: sentiment lexicons are domain-specific, and applying a finance lexicon to fiction under-counts sentiment and misses most of the emotional texture of the text.

The broader takeaway is the same one the book makes for Jane Austen: the choice of lexicon materially changes the story you tell about a text, so comparing multiple lexicons is more informative than trusting any single one.